

1 Introduction

Relative position refers to the positioning of a point with respect to another point or points, and is commonly used in geodesy to determine spatial relationships. It includes techniques such as angle and distance measurements that are differential in nature and can be computed on a reference ellipsoid or mapping plane. [5]

We can use GNSS systems, such as GPS, to compute the precise position of the receivers. This is because GNSS satellites transmit signals that allow us to accurately estimate the position of the receivers. Relative positioning is a technique we use to compute these positions, it involves the use of two receivers, one receiver is stationary and its exact position is known, the other is roving and its position needs to be estimated.

In contrast to relative positioning, absolute positioning determines the position of a receiver directly with respect to the Earth's surface. This is the main difference from relative positioning, where the position of a receiver is determined with respect to another known point.

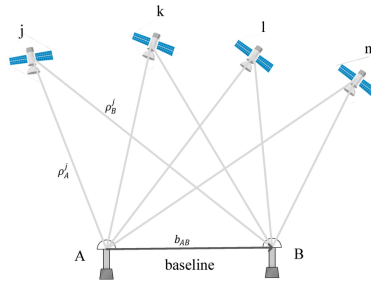


Figure 1: Relative positioning [6].

The GNSS satellites continuously transmit navigation signals at two or more frequencies in the L-band. These signals contain ranging codes and navigation data that allow users to compute the travel time from the satellite to the receiver and the satellite coordinates at any epoch. [2]

The GNSS signals contain ranging codes to allow users to compute the travel time ΔT . The receiver determines ΔT by correlating the received code (P) from the satellite with a replica of it generated in the receiver, moving this replica in time (ΔT) until the maximum correlation is obtained. This measurement $c\Delta T$ is the pseudorange. It is an "apparent range" between the satellite and the receiver that does not match the geometric distance due to factors such as synchronization errors between receiver and satellite clocks. The travel time between transmission and reception is obtained as the difference of the times measured between the clocks or time scales in the satellite and in the receiver. [1]

In addition to the code, the carrier phase itself is used to obtain a measure of the apparent distance between the satellite and the receiver. These carrier phase measurements are much more accurate than the code measurements, but they are ambiguous by an unknown integer number of wavelengths (λN). This ambiguity changes arbitrarily every time the receiver loses lock on the signal, causing jumps or range discontinuities [1]. If there is an interruption in the signal that causes the receiver to lose lock, the phase will exhibit a discontinuity of an integer number of cycles ("cycle slip"). This discontinuity may be only a few cycles, due to a low signal-to-noise ratio, or it may be thousands of cycles, as may occur if the satellite is blocked at the receiver site. Initial processing of the phase data is often performed using time differences of doubly differenced phase ("triple differences" or

"Doppler" observations) to provide a preliminary estimate of station or orbital parameters in the presence of cycle slips [4]. Carrier phase measurements are ambiguous by an integer number of wavelengths. Techniques like the Least-squares Ambiguity Decorrelation Adjustment (LAMBDA) method are used to resolve these ambiguities. The LAMBDA method has been widely used in GNSS for fixing integer ambiguities. It can also solve any integer least squares (ILS) problem arising from other applications. It consists of three steps including a float solution generation, a decorrelation of double difference measurements, and an integer ambiguity search.

By combining code observations and carrier phase measurements, we can compute the unknown position using a mathematical model based on the difference of the measurements at the stationary and roving receivers. By leveraging differencing techniques and resolving integer ambiguities, it is possible to determine the rover's position with centimeter-level accuracy.

Pseudoranges are useful for initial position estimates and resolving clock biases. However, they are less precise compared to phase observations due to higher noise levels, that is why carrier phases are crucial for achieving high-precision positioning. They are used to refine the position estimates obtained from pseudoranges.

The differential processes involved in the computation of the positions are Single Differencing and Double Differencing. While single differencing involves subtracting the observations from the base station from those of the rover for each satellite allowing the elimination of satellite clock errors, double differencing involves subtracting the single differences of one satellite from another. This process removes receiver clock errors and reduces other common errors.

The goal of this assignment will be the computation of the rover coordinates, in order to achieve this we will formulate the observation equations based on the differenced pseudoranges and carrier phases. Then by the LAMBDA method we will solve for integer Ambiguities. Once the ambiguities are resolved, the precise position of the rover can be calculated. We will compute the standard deviations from the receiver coordinates, computed using the best set of integer values of ambiguities. We will perform a comparison between the float and integer ambiguities. And by ambiguity validation we will analyze the fixed ambiguities.

2 Methodology

The data given consist in in the code and phase observations from two receivers (the reference and the rover). We will also have the coordinates X,Y,Z, clock errors and topocentric distances of the eight satellites available at the moment of taking the measurements. We will also have the coordinates of the reference and the approximate coordinates of the rover.

For the computation of the relative positioning of the rover we will follow the mathematical processes included in [3].

- **Step 1: Synchronize observables from both receivers**

In Step 1, we synchronize the observables from both receivers to correct for the receiver clock differences and satellite signal travel time offsets before computing single differences.

This synchronization is made in both code and phase measurements for the rover and the reference following the equation:

$$P_A^s = P_A^s(t_A) = P_A^s(t_A) + \dot{P}_A^s(t_A) \frac{\delta \hat{t}_A}{c}$$

$$\lambda \phi_A^s = \lambda \phi_A^s(t_A) = \lambda \phi_A^s(t_A) + \dot{\phi}_A^s(t_A) \frac{\delta \hat{t}_A}{c}$$

The first equation refers to the pseudoranges, given by $P_A^s(t_A)$ and the second one refers to the phase observations given by $\phi_A^s(t_A)$. The term $\dot{P}_A^s(t_A)$ represents the topocentric range rates, i.e., how fast the distance to the satellite is changing. The receivers clock error is given by $\delta \hat{t}_A$, since this is given in meters we should divide by the speed of light c to transform it into seconds.

The subindex A could be *Ref* referring to the reference or *Rov* that represents the rover.

- **Step 2: Compute single differences**

In this step, we compute single differences for code and phase measurements to eliminate satellite clock errors by differencing observations from both receivers for the same satellite.

$$P_{Ref,Rov}^s = P_{Rov}^s - P_{Ref}^s$$

$$\lambda \phi_{Ref,Rov}^s = \lambda \phi_{Rov}^s - \lambda \phi_{Ref}^s$$

where P_{Rov}^s , P_{Ref}^s , ϕ_{Rov}^s , and ϕ_{Ref}^s are the synchronization rover and reference pseudorange measurements and the synchronization rover and reference carrier phase measurements, respectively.

- **Step 3: Compute double differences**

In this step, we eliminate receiver clock errors by differencing single differences of two satellites. We will take the first satellite as the reference, and we will compute double differences for the other satellites. This gives us relative measurement that is free from both satellite and receiver clock errors.

$$P_{Ref,Rov}^{st} = P_{Ref,Rov}^t - P_{Ref,Rov}^s$$

$$\lambda \phi_{Ref,Rov}^{st} = \lambda \phi_{Ref,Rov}^t - \lambda \phi_{Ref,Rov}^s$$

Here the super-index t refers to the satellite 1, and s to the rest of the satellites. We use the single differences previously computed.

- **Step 4: Compute coefficients a_X, a_Y, a_Z and single-difference distances $\rho_{AB,0}^s$**

We derive partial derivatives of range with respect to the rover's coordinates.

$$\frac{\partial \rho_{Rov,0}^s(t_{Rov})}{\partial X_{Rov,0}} = a_X^s = -\frac{X^s - X_{Rov,0}}{\rho_{Rov,0}^s}$$

$$\frac{\partial \rho_{Rov,0}^s(t_{Rov})}{\partial Y_{Rov,0}} = a_Y^s = -\frac{Y^s - Y_{Rov,0}}{\rho_{Rov,0}^s}$$

$$\frac{\partial \rho_{Rov,0}^s(t_{Rov})}{\partial Z_{Rov,0}} = a_Z^s = -\frac{Z^s - Z_{Rov,0}}{\rho_{Rov,0}^s}$$

Where $\rho_{Rov,0}^s$ is the topocentric distance for the rover given in the data. The coordinates X^s, Y^s, Z^s are the satellites coordinates and $X_{Rov,0}, Y_{Rov,0}, Z_{Rov,0}$ are the approximate coordinates of the rover.

The topocentric distance single difference is computed as:

$$rho_{Ref,Rov,0}^s = rho_{Ref,0}^s - rho_{Rov,0}^s$$

- **Step 5: Compute double-differenced coefficients and $\rho_{AB,0}^{st}$**

In this step we find changes in the direction vectors between satellites relative to the reference satellite. We will do this by computing the double differences with the coefficients computed in the previous step.

$$a_{X_{Rov}}^{st} = a_{X_{Rov}}^t - a_{X_{Rov}}^s$$

$$a_{Y_{Rov}}^{st} = a_{Y_{Rov}}^t - a_{Y_{Rov}}^s$$

$$a_{Z_{Rov}}^{st} = a_{Z_{Rov}}^t - a_{Z_{Rov}}^s$$

The super-index t refers to the satellite 1, and s to the rest of the satellites.

We compute the double difference topocentric distance as

$$\rho_{Ref,Rov,0}^{st} = \rho_{Ref,Rov,0}^t - \rho_{Ref,Rov,0}^s$$

- **Step 6: Construct matrix **A** and vector **L****

We create the design matrix **A** that contains the double differences between the satellites with respect to the first satellite as computed in the previous step.

$$\mathbf{A} = \begin{bmatrix} a_{X_{Rov}}^{1,2} & a_{Y_{Rov}}^{1,2} & a_{Z_{Rov}}^{1,2} & \lambda_1 & 0 & \cdots & 0 \\ a_{X_{Rov}}^{1,3} & a_{Y_{Rov}}^{1,3} & a_{Z_{Rov}}^{1,3} & 0 & \lambda_1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{X_{Rov}}^{1,8} & a_{Y_{Rov}}^{1,8} & a_{Z_{Rov}}^{1,8} & 0 & 0 & \cdots & \lambda_1 \\ a_{X_{Rov}}^{1,2} & a_{Y_{Rov}}^{1,2} & a_{Z_{Rov}}^{1,2} & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{X_{Rov}}^{1,8} & a_{Y_{Rov}}^{1,8} & a_{Z_{Rov}}^{1,8} & 0 & 0 & \cdots & 0 \end{bmatrix}$$

The vector $\mathbf{L}$ is the observation vector and it contains the observed double differences phase and code residuals.

$$\mathbf{L} = \begin{bmatrix} \lambda\phi_{Ref,Rov}^{1,2} - \rho_{Ref,Rov,0}^{1,2} \\ \vdots \\ \lambda\phi_{Ref,Rov}^{1,8} - \rho_{Ref,Rov,0}^{1,8} \\ p_{Ref,Rov,1}^{1,2} - \rho_{Ref,Rov,0}^{1,2} \\ \vdots \\ p_{Ref,Rov,1}^{1,8} - \rho_{Ref,Rov,0}^{1,8} \end{bmatrix}$$

Now that we have we should perform one least square iteration to determine the vector $\mathbf{x}$ that contains position corrections and float ambiguities.

$$\mathbf{x} = \left[\Delta X \Delta Y \Delta Z N_{Ref,Rov}^{1,2} \dots N_{Ref,Rov}^{1,8} \right]^T$$

We should solve

$$\mathbf{L} = \mathbf{A}\mathbf{x} \rightarrow \mathbf{x} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \mathbf{L}$$

Where $\mathbf{P}$ is the weight matrix defined as

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_\Phi & 0 \\ 0 & \mathbf{P}_P \end{bmatrix}$$

$$\mathbf{P}_i = \frac{1}{2\sigma_i^2(n_{dd} + 1)} \begin{bmatrix} n_{dd} & -1 & -1 & \dots \\ -1 & n_{dd} & -1 & \dots \\ -1 & & \ddots & \\ \vdots & \dots & & n_{dd} \end{bmatrix}, \quad i = \Phi, P$$

And, n_{dd} is the number of double difference observations, in this case 7. The standard deviation of undifferenced phase and code are $\sigma_\phi = 2mm$ and $\sigma_P = 0.3m$.

- **Step 7: Update the ROV position**

We update the estimated position using the correction from LSQ solution $\mathbf{x}$.

$$X_{Rov} = X_{Rov,0} + \Delta X$$

$$Y_{Rov} = Y_{Rov,0} + \Delta Y$$

$$Z_{Rov} = Z_{Rov,0} + \Delta Z$$

Here $X_{Rov,0}, Y_{Rov,0}, Z_{Rov,0}$ are the approximate coordinates of the river given, and $\Delta X, \Delta Y, \Delta Z$ are the position corrections given by $\mathbf{x}$.

- **Step 8: Compute the variance-covariance matrix $Q_{\hat{a}}$**

From the vector $\mathbf{x}$ we extract the ambiguities given by

$$\hat{a} = \left[N_{Ref,Rov}^{1,2} \dots N_{Ref,Rov}^{1,8} \right]^T,$$

which are the vector $\mathbf{x}$ elements from the fourth until the end.

We will compute the LSQ covariance matrix as

$$Q_x = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1},$$

from this matrix we will extract the part that belongs to the ambiguity, this gives us the matrix $Q_{\hat{a}}$, which is the matrix with rows and columns that form the 4th to the end of Q_x .

- **Step 9: Run the LAMBDA method**

We will use the LAMBDA function provided by TU Delft, and we will use as our input the float ambiguity estimates $\hat{a}$ and the variance-covariance matrix $Q_{\hat{a}}$ computed in the previous step. Our output from the LAMBDA function will be the integer (fixed) ambiguities.

- **Step 10: Use fixed ambiguities to update $\mathbf{L}$**

In this step we will modify the vector $\mathbf{L}$. Now, the vector will be reduce to have dimension $n_{dd} \times 1$ and we focus in the phase observations.

Now since the ambiguities are known, we should include them in the computation of this vector. The best set of fixed ambiguities are saved in the variable $afixed_1$ obtained from the LAMBDA function, while the second best set of fixed ambiguities are saved in $afixed_2$.

$$\mathbf{L}_{best} = \begin{bmatrix} \lambda\phi_{Ref,Rov}^{1,2} - \rho_{Ref,Rov,0}^{1,2} - \lambda afixed_{1,2} \\ \vdots \\ \lambda\phi_{Ref,Rov}^{1,8} - \rho_{Ref,Rov,0}^{1,8} - \lambda afixed_{1,8} \end{bmatrix}$$

We can also compute the vector $\mathbf{L}_{secondbest}$ for the second best set of ambiguities following the same procedure but changing $afixed_1$ by $afixed_2$.

- **Step 11: Rebuild $\mathbf{A}$ matrix**

Reduce $\mathbf{A}$ matrix to only contain geometric coefficients a_X , a_Y and a_Z . This is matrix of dimension $n_{dd} \times 3$, and we do not included the ambiguity columns in the previous matrix $\mathbf{A}$. We now solve only for position with fixed ambiguities.

- **Step 12: Rebuild $\mathbf{P}$ matrix**

We build the new weight matrix $\mathbf{P}$, taking from the matrix $\mathbf{P}$ previously computed only the part that corresponds to the phase observations.

$$\mathbf{P}_{phi} = \frac{1}{2\sigma_{phi}^2(n_{dd} + 1)} \begin{bmatrix} n_{dd} & -1 & -1 & \dots \\ -1 & n_{dd} & -1 & \dots \\ -1 & & \ddots & \\ \vdots & \dots & & n_{dd} \end{bmatrix}$$

- **Step 13: Resolve for $\mathbf{x}$ again using updated $\mathbf{L}$, $\mathbf{A}$, and $\mathbf{P}$ matrices**

Given the new values of $\mathbf{L}$, $\mathbf{A}$ and $\mathbf{P}$ computed, we can solve the system to obtain the values of $\mathbf{x}$, that can be used as in **Step 7** to actualize the rover coordinates.

$$\mathbf{L} = \mathbf{A}\mathbf{x} \rightarrow \mathbf{x} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \mathbf{L}$$

$$X_{Rov} = X_{Rov,0} + \Delta X$$

$$Y_{Rov} = Y_{Rov,0} + \Delta Y$$

$$Z_{Rov} = Z_{Rov,0} + \Delta Z$$

We obtain a high-precision rover position estimate using the best set of fixed integer ambiguities.

We use this process to compute the position correction vector $\mathbf{x}$ for the best set of fixed ambiguities and the second best set. The computation will differ in the selection of the vector $\mathbf{L}$. We save the results in $\mathbf{x}_{best}$ and $\mathbf{x}_{secondbest}$, respectively.

- **Step 14: Perform ambiguity validation**

Given the the $\mathbf{L}$, $\mathbf{x}$ and $\mathbf{A}$ computed for the best and second best set of fixed ambiguities we can compute the residual vector as

$$\mathbf{v} = \mathbf{L} - \mathbf{A}\mathbf{x}.$$

With the residual vectors $\mathbf{v}_{best}$ and $\mathbf{v}_{secondbest}$, we can perform the ambiguity validation test by computing the ratio

$$\frac{(\mathbf{v}^T \mathbf{P} \mathbf{v})_{secondbest}}{(\mathbf{v}^T \mathbf{P} \mathbf{v})_{best}} > \text{critical value}.$$

This ratio is then compared with a critical value that generally can be set from the number of degrees of freedom in the adjustment or set to a fixed value. In this case we take the critical value to be 3.

This process has been programmed in Matlab.

The results from the intermediate steps 1-7 has been compared with the ones provided in the file "Diff AssignmentL1data.xlsx", and we where able to see the high accuracy of the computed results with respect to the ones given in the file.

3 Results

After fixing the integer ambiguities in **Step 9-12**, we have compute in **Step 13** the final coordinates of the rover by taking the approximate coordinates given at first and adapting them by the corrections computed also in this step from the vector $\mathbf{x}$. The standard deviations of each coordinate were extracted from the variance-covariance matrix computed in the LAMBDA function.

We include the final rover coordinates and the computed standard deviations:

- **X:** 3101687.3968 $\pm$ 0.2166 m
- **Y:** 1012242.3985 $\pm$ 0.1678 m
- **Z:** 5462199.2989 $\pm$ 0.1762 m

These results demonstrate centimeter-level accuracy, highlighting the effectiveness of ambiguity fixing.

In **Step 6** we have compute the float ambiguities from the initial least squares solution. In Table 1 we add the comparison between the float ambiguities and the fixed integer values obtained via the LAMBDA method in **Step 9**.

	Float Ambiguity	Fixed Ambiguity	Difference
1	-1.013991	-1	-0.013991
2	-6.726760	-7	0.273240
3	-8.640985	-9	0.359015
4	-13.558093	-13	-0.558093
5	-19.987093	-20	0.012907
6	-21.867676	-22	0.132324
7	-25.184002	-26	0.815998

Table 1: Float vs Fixed Ambiguities

From the comparison between the float ambiguities and the fixed ambiguities we can see that in general the difference between both ambiguities is small, however we can see that the difference between the ambiguities for the last satellite takes a higher value, suggesting the presence of increased noise or potential model inaccuracies affecting this specific satellite-receiver geometry.

To validate the reliability of the fixed ambiguity, in **Step 14** we have compute the ratio to perform the ambiguity validation.

$$\frac{(\mathbf{v}^T \mathbf{P} \mathbf{v})_{secondbest}}{(\mathbf{v}^T \mathbf{P} \mathbf{v})_{best}} > 3$$

In this case we have obtain that the ratio get a value of 22.23, which is higher than 3. Since the ratio exceeds the threshold given, the ambiguity fix is accepted, indicating that the integer solution is statistically reliable.

4 Analysis and Discussion

In the task we computed the rover position from the given phase and pseudorange observations of the rover and the reference. We computed single and double differences and computed the ambiguities. From these we computed the fixed ambiguities from the phase observations.

The fixed ambiguities transformed the phase observations into highly accurate measurements, improving the geometry of the observation system and reducing the effect of measurement noise. This transition from a float to a fixed solution is essential for achieving centimeter-level positioning in high-precision GNSS applications.

From the results obtained, we have noticed the small standard deviation of the computed position, which means that the process of ambiguity fixing has an important effect on the position accuracy.

While most of the float ambiguities were within ± 0.5 intervals of the nearest integer, indicating a high consistency with the fixed values, one ambiguity showed a significantly larger difference of about 0.8.

This deviation suggests the presence of increased noise or potential model inaccuracies affecting this particular satellite receiver geometry. Nevertheless, the ambiguity set was accepted based on the overall ratio test, which yielded a value of 22.23, exceeding the confidence threshold of 3.

It is important to note that the LAMBDA method minimizes the sum of the squared residuals over all ambiguities. This means that a single ambiguity does not necessarily invalidate the solution if the overall residual is still minimized and the solution is statistically valid.

References

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